

Khushboo Rani

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PROFILE

Computer architecture researcher with 4+ years of postdoctoral research experience across three institutions (Virginia Tech, University of Florida, NUS Singapore). Co-PI on a funded ONR project (\$986,606) on microcode-level security enforcement. Published in IEEE TCAD, IEEE TVLSI, DAC, and ICCD. Available immediately for research positions in secure processor architecture and AI systems security.

RESEARCH INTEREST

My research focuses on secure processor design and architecture for AI systems — spanning microcode-level memory safety enforcement, side-channel attack mitigation (including Spectre-class transient execution vulnerabilities), and on-chip interconnect design for AI accelerators. I build deployable defenses on commodity hardware, enforcing memory compartmentalization with under 3% overhead on production workloads without requiring ISA extensions or application changes. My goal is to develop processor architectures that are fundamentally trustworthy as AI systems scale across heterogeneous compute fabrics.

HONORS AND AWARDS

- Received Best Paper Award in ISED 2018
- Nominated for the Best Paper Award in VLSID 2019
- Received Grant from IARCS/ACM India for attending DAC 2020 conference in San Francisco, USA
- DAC Young Fellow 2020

EDUCATION

Indian Institute of Technology Guwahati, India

July 2012 – Jan 2021

Ph.D. Computer Science & Engineering

Advisor: Prof. Hemangee Kapoor

Thesis Title: LongLiveNoC: Wear Levelling, Write Reduction and Selective VC allocation for Long-lasting Dark Silicon aware NoC Interconnects

Cochin University of Science and Technology, India

Aug 2005 – June 2009

B. Tech. Computer Science & Engineering

EXPERIENCE

Department of ECE, Virginia Tech, USA | *Postdoctoral Associate*

May 2023 – Aug 2026

Advisor: Prof. Binoy Ravindran

Co-PI on a funded Office of Naval Research (ONR) project (TrustedCPU, \$986,606) advancing microcode-level security enforcement on commodity x86-64 processors. Completed full control-flow enforcement implementation at the microcode layer and developed memory isolation mechanisms for spatial and temporal safety without ISA extensions or binary modifications. Contributed to DARPA/V-SPELLS and DARPA/BARC research on the IOCS framework for input-aware compartmentalization, enforcing fine-grained memory isolation with under 3% overhead on production workloads. Mentored 2 master's students on architecture and security research.

Department of CISE, University of Florida, USA | *Postdoctoral Associate*

May 2022 – April 2023

Advisor: Prof. Prabhat Mishra

Served as lab lead for a team of 5-6 researchers on the DARPA/Synopsys Automated Implementation of Secure Silicon (AISS) program. Delivered a malicious hardware implant detection tool and cryptographic IP implementations for

supply-chain trust during SoC design and fabrication. Results published at ICCD 2022 and presented at GOMACTech 2023. Mentored 1 PhD student and 2 undergraduate researchers within a one-year appointment.

School of Computing, NUS, Singapore | *Research Fellow*

May 2021 – Dec 2021

Advisor: Dr. Trevor Carlson

Contributed to hardware/software co-design research on mitigating Spectre-class transient execution attacks through compiler-assisted secure execution mechanisms and gem5-based microarchitectural modeling. Work conducted remotely during COVID-19 travel restrictions prior to joining University of Florida.

PUBLICATIONS

JOURNAL

- Hansika Weerasena, Pan Zhixin, **Khushboo Rani** and Prabhat Mishra, “ML-based Flow Correlation Attack on Anonymous Routing and Lightweight Countermeasures for NoC-based SoCs,” manuscript developed during postdoctoral research at the University of Florida (2022–2023).
- **Khushboo Rani** and Hemangee K. Kapoor, “Investigating Frequency Scaling, Non-Volatile, and Hybrid Memory Technologies for On-Chip Routers to Support the Era of Dark Silicon.” In *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems* 40, no. 4 (2020): 633-645. DOI: 10.1109/TCAD.2020.3007555
- **Khushboo Rani** and Hemangee K. Kapoor, “Write Variation Aware Buffer Assignment for Improved Lifetime of Non-Volatile Buffers in On-Chip Interconnects.” In *IEEE Transactions on Very Large Scale Integration (VLSI) Systems* 27, no. 9 (2019): 2191-2204. DOI: 10.1109/TVLSI.2019.2922471
- **Khushboo Rani** and Hemangee K. Kapoor, “Write-Variation Aware Alternatives to Replace SRAM Buffers with Non-Volatile Buffers in On-Chip Interconnects.” In *IET Computers and Digital Techniques* 13, no. 6 (2019): 481-492. DOI: 10.1049/iet-cdt.2019.0039

CONFERENCE

- **Khushboo Rani** and Binoy Ravindran, “Microcode-based Compartmentalization Enforcement for Fine-Grained Software Isolation,” manuscript in preparation for submission.
This work is supported by the Office of Naval Research (ONR) TrustedCPU project.
- Jae-Won Jang, Xiaoguang Wang, **Khushboo Rani**, David Narvaez, and Binoy Ravindran, “IOCS: Input-Oriented Object Compartmentalization for Memory Damage Containment,” manuscript in preparation for submission.
This work is supported by DARPA V-SPELLS and DARPA BARC programs.
- **Khushboo Rani**, Hansika Weerasena, Stephen A. Butler, Subodha Charles, and Prabhat Mishra, “Modeling and Exploration of Gain Competition Attacks in Optical Network-on-Chip Architectures.” Available: <https://arxiv.org/abs/2303.01550>, 2023. DOI: 10.48550/arXiv.2303.01550
- Aruna Jayasena, **Khushboo Rani**, and Prabhat Mishra, “Efficient Finite State Machine Encoding for Defending Against Laser Fault Injection Attacks.” In *2022 IEEE 40th International Conference on Computer Design (ICCD)*, pp. 247-254. IEEE, 2022. DOI: 10.1109/ICCD56317.2022.00044
This work was supported by the DARPA/Synopsys AISS program.
- **Khushboo Rani**, Sukarn Agarwal, and Hemangee K. Kapoor, “DidaSel: Dirty data based Selection of VC for effective utilization of NVM Buffers in On-Chip Interconnects.” In *Proceedings of the ACM/IEEE International Symposium on Low Power Electronics and Design*, pp. 151-156. 2020. DOI: 10.1145/3370748.3406565
- **Khushboo Rani** and Hemangee K. Kapoor, “ZENCO: Zero-bytes based ENCOding for Non-Volatile Buffers in On-Chip Interconnects.” In *2020 57th ACM/IEEE Design Automation Conference (DAC)*, pp. 1-6. IEEE, 2020. DOI: 10.1109/DAC18072.2020.9218620

- **Khushboo Rani** and Hemangee K. Kapoor, “Write Variation Aware Non-Volatile Buffers for On-Chip Interconnects.” In *2019 32nd International Conference on VLSI Design and 18th International Conference on Embedded Systems (VLSID)*, pp. 7-12. IEEE, 2019. (**Best Paper Candidate**). DOI: 10.1109/VLSID.2019.00020
- **Khushboo Rani**, Sukarn Agarwal, and Hemangee K. Kapoor, “Non-blocking Gated Buffers for Energy Efficient on-chip Interconnects in the era of Dark Silicon.” In *2018 8th International Symposium on Embedded Computing and System Design (ISED)*, pp. 74-79. IEEE, 2018. (**Best Paper Award**). DOI: 10.1109/ISED.2018.8704027

PROPOSALS

PROPOSALS FUNDED

- “TrustedCPU: Formally verified microcode customization for security exploit mitigation,” Source: ONR, Dr. Binoy Ravindran (PI), Dr. Freek Verbeek (Co-PI), Dr. Khushboo Rani (Co-PI), Dr. Ruslan Nikolaev (Co-PI), Total amount: \$986,606.00

PROPOSALS SUBMITTED

- “Seal: Fully Automatic, Formally-verified Memory Isolation of Legacy Software with Application to Linux Device Drivers,” Source: ONR, Dr. Binoy Ravindran (PI), Dr. Freek Verbeek (Co-PI), Dr. Jonas Haglund (Co-PI), Dr. Khushboo Rani (Co-PI), Total amount: \$948,115.00

POSTERS

- **Khushboo Rani**, Emma Andrews, Aruna Jayasena and Prabhat Mishra, “Defending Elliptic Curve Cryptography against Laser Fault Injection Attacks”, In *GOMACTech*, 2023.

PHD FORUM

- VLSI Design 2020, Bangalore, India.
- DATE 2021, Virtual Conference & Exhibition.
- ASPDAC 2021, Virtual Conference.

TEACHING ASSISTANT

INDIAN INSTITUTE OF TECHNOLOGY GUWAHATI, INDIA

- * Computer Architecture (CS223 : UG)
- * Parallel Computer Architecture (CS527 : UG+PG)
- * Computational Geometry (CS502 : UG+PG)
- * Hardware Laboratory (CS224 : UG)
- * System Software Lab (CS : UG)
- * Theory of computation
- * Introduction to Computing (CS101 : UG)

INDUSTRY EXPERIENCE

Wipro Technologies, Bangalore India | *Project Engineer*

May 2010 – July 2012

Worked on two projects: (i) Nokia app development (ii) NetApp file system development and maintenance

SKILLS

Languages: C/C++ (expert), Python (scripting — data extraction, automation), Shell scripting (benchmark automation)

Operating Systems/Platforms: Linux (primary), Unix, Docker (containerized simulation environments)

Architecture Simulation: GEM5, Garnet, Noxim, Booksim, DSENT (expert)

Architecture Modeling Tools: Orion, McPAT (used in publications), CACTI, NVSIM (NVM buffer configuration), HotSpot

Security & Low-level Tools: x86 Microcode (expert), HOL4 (x86 MMU modeling), Intel PIN (working knowledge)

Hardware: FPGA (basic)

Other: LaTeX (expert — papers, CV, statements)